

8.5 Quiz: Rational Functions and Transformations — Markscheme

Version (a)

1. $f(x) = \frac{2}{x-3} + 1.$

(a) Vertical asymptote: $x = 3.$ A1

Horizontal asymptote: $y = 1.$ A1

(b) A vertical stretch with scale factor 2, A1

a horizontal translation 3 units to the right, A1

a vertical translation 1 unit up. A1

Accept the two translations combined as a translation by the vector $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ (A1A1). The stretch and the translations may be described in either order.

(c) $f(0) = \frac{2}{-3} + 1 = \frac{1}{3},$ so $(0, \frac{1}{3}).$ A1

Accept $y = \frac{1}{3}.$

(d) $\frac{2}{x-3} + 1 = 0,$ $\frac{2}{x-3} = -1,$ $x-3 = -2$ M1

$x = 1,$ so $(1, 0).$ A1

Accept $x = 1.$

2. $g(x) = \frac{2x+1}{x-1},$ $x \neq 1.$

(a) Vertical asymptote: $x = 1.$ A1

Horizontal asymptote: $y = 2$ (ratio of leading coefficients $= \frac{2}{1}$). A1

(b) x -intercept: $2x+1=0 \Rightarrow x = -\frac{1}{2},$ so $(-\frac{1}{2}, 0).$ A1

y -intercept: $g(0) = \frac{1}{-1} = -1,$ so $(0, -1).$ A1

Accept $x = -\frac{1}{2}$ and $y = -1.$

(c) Let $y = \frac{2x+1}{x-1}$ and swap x and $y:$ $x = \frac{2y+1}{y-1}$ M1

$x(y-1) = 2y+1,$ $xy-2y = x+1,$ $y(x-2) = x+1$ A1

$g^{-1}(x) = \frac{x+1}{x-2}$ A1

(d) Domain of g^{-1} : $x \in \mathbb{R}, x \neq 2$. A1

FT: award A1 for $x \neq$ (the candidate's horizontal-asymptote value of g from (a)).

Version (b)

1. $f(x) = \frac{3}{x+2} - 1$.

(a) Vertical asymptote: $x = -2$. A1

Horizontal asymptote: $y = -1$. A1

(b) A vertical stretch with scale factor 3, A1

a horizontal translation 2 units to the left, A1

a vertical translation 1 unit down. A1

Accept the two translations combined as a translation by the vector $\begin{pmatrix} -2 \\ -1 \end{pmatrix}$ (A1A1). The stretch and the translations may be described in either order.

(c) $f(0) = \frac{3}{2} - 1 = \frac{1}{2}$, so $(0, \frac{1}{2})$. A1

Accept $y = \frac{1}{2}$.

(d) $\frac{3}{x+2} - 1 = 0$, $\frac{3}{x+2} = 1$, $x+2 = 3$ M1

$x = 1$, so $(1, 0)$. A1

Accept $x = 1$.

2. $g(x) = \frac{3x-2}{x+1}$, $x \neq -1$.

(a) Vertical asymptote: $x = -1$. A1

Horizontal asymptote: $y = 3$ (ratio of leading coefficients = $\frac{3}{1}$). A1

(b) x -intercept: $3x - 2 = 0 \Rightarrow x = \frac{2}{3}$, so $(\frac{2}{3}, 0)$. A1

y -intercept: $g(0) = \frac{-2}{1} = -2$, so $(0, -2)$. A1

Accept $x = \frac{2}{3}$ and $y = -2$.

(c) Let $y = \frac{3x-2}{x+1}$ and swap x and y : $x = \frac{3y-2}{y+1}$ M1

$x(y+1) = 3y-2$, $xy - 3y = -x - 2$, $y(x-3) = -(x+2)$ A1

$$g^{-1}(x) = -\frac{x+2}{x-3} \quad \mathbf{A1}$$

Accept the equivalent form $\frac{x+2}{3-x}$.

(d) Domain of g^{-1} : $x \in \mathbb{R}, x \neq 3$. **A1**

FT: award A1 for $x \neq$ (the candidate's horizontal-asymptote value of g from (a)).

Version (c)

1. $f(x) = \frac{4}{x-1} + 2$.

(a) Vertical asymptote: $x = 1$. **A1**

Horizontal asymptote: $y = 2$. **A1**

(b) A vertical stretch with scale factor 4, **A1**

a horizontal translation 1 unit to the right, **A1**

a vertical translation 2 units up. **A1**

Accept the two translations combined as a translation by the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ (A1A1). The stretch and the translations may be described in either order.

(c) $f(0) = \frac{4}{-1} + 2 = -2$, so $(0, -2)$. **A1**

Accept $y = -2$.

(d) $\frac{4}{x-1} + 2 = 0$, $\frac{4}{x-1} = -2$, $x-1 = -2$ **M1**

$x = -1$, so $(-1, 0)$. **A1**

Accept $x = -1$.

2. $g(x) = \frac{4x+3}{x-2}$, $x \neq 2$.

(a) Vertical asymptote: $x = 2$. **A1**

Horizontal asymptote: $y = 4$ (ratio of leading coefficients $= \frac{4}{1}$). **A1**

(b) x -intercept: $4x+3=0 \Rightarrow x = -\frac{3}{4}$, so $(-\frac{3}{4}, 0)$. **A1**

y -intercept: $g(0) = \frac{3}{-2} = -\frac{3}{2}$, so $(0, -\frac{3}{2})$. **A1**

Accept $x = -\frac{3}{4}$ and $y = -\frac{3}{2}$.

(c) Let $y = \frac{4x+3}{x-2}$ and swap x and y : $x = \frac{4y+3}{y-2}$ **M1**

$x(y-2) = 4y+3$, $xy-4y = 2x+3$, $y(x-4) = 2x+3$ **A1**

$g^{-1}(x) = \frac{2x+3}{x-4}$ **A1**

(d) Domain of g^{-1} : $x \in \mathbb{R}, x \neq 4$. **A1**

FT: award A1 for $x \neq$ (the candidate's horizontal-asymptote value of g from (a)).